

Selection Sort: Sabse Chhota Dhundo!

1. Selection Sort Kya Hai?

Maan lo ki ek naya shipment aaya hai brass decorative lights ka, aur sab alag-alag sizes ke hain. **Selection Sort** mein hum inhe kaise arrange karenge?

- Saari lights ko dekhte hain aur **sabse chhoti (smallest)** light dhundhte hain.
- Usko utha kar display mein **sabse pehli jagah** par rakh dete hain.
- Phir bachi hui lights mein se sabse chhoti dhundhte hain aur usko **dusri jagah** par rakh dete hain.

Aise hi agar gym mein dumbbells arrange karne ho, toh sabse halka dumbbell dhundh kar sabse pehle rakha jayega! Yahi simple sa logic Selection Sort ka hai. ✨

2. Step-by-Step Detective Work

Jaise *Feluda* ya *Byomkesh Bakshi* ek-ek clue ko dhyan se dekh kar asli sach ko dhundhte hain, waise hi hum array mein se ek-ek element ko check karke **Minimum (sabse chhota)** number dhundhte hain!

Numbers: 64, 25, 32, 20, 15

Step 1 (First Iteration): Pura array dekho. Sabse chhota kon hai? **15!**



Ab 15 ko sabse pehli jagah (64 ki jagah) par bhej do aur 64 ko 15 ki jagah par bhej do (Swapping!).



Step 2 (Second Iteration): Ab 15 toh apni jagah par set ho gaya. Bache hue numbers (25, 32, 20, 64) mein fir se detective ban kar sabse chhota dhundo. Wo hai **20!**

Toh 20 ko bache hue array ki pehli jagah (yani 25 ki jagah) par swap kar do.



Aise hi hum baar-baar sabse chhota number dhundhte hain aur use aage lagate jate hain, jab tak saare numbers badhte kram (increasing order) mein sort na ho jayein!



3. Python Code Ka Logic

```
def selection_sort(A):  
    n = len(A)  
    # i batata hai ki hume ab array mein kahan se check karna  
    start karna hai  
    for i in range(n):  
        # Pehle maan lo ki i wali jagah pe hi sabse chhota  
        number rakha hai  
        min_index = i  
  
        # Ab bache hue array mein asli chhota number dhundo  
        (Detective mode ON!)  
        for j in range(i + 1, n):  
            if A[j] < A[min_index]:
```

```
        min_index = j # Naya sabse chhota number mil
gaya!

        # Jab sabse chhota element mil jaye, toh usko 'i' wali
jagah se SWAP kar do! ↻
        A[i], A[min_index] = A[min_index], A[i]

# Code Test Karte Hain!
arr = [64, 25, 32, 20, 15]
selection_sort(arr)
print("Sorted Array:", arr)
```

🌟 Important Points to Remember!

- **Main Logic:** Humesha *Minimum* element ko dhundo aur usko uski sahi jagah par swap kar do.
- **Loops:** Bubble sort ki tarah hi isme bhi do (2) loops use hote hain (Ek kahan rakhna hai uske liye, aur dusra sabse chhota element dhundhne ke liye).
- **Time Complexity:** Iski Time Complexity bhi $O(n^2)$ hoti hai. Kyunki outer loop n times chalta hai aur inner loop bhi approx n times chalta hai.